

V Mahesh

Data Engineer

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SUMMARY

Data Engineer with 5+ years of experience designing and building scalable ETL/ELT pipelines in financial services environments. Skilled in big data processing using Databricks, PySpark, Spark, and distributed systems, with hands-on experience in Azure and working knowledge of AWS (S3, Glue). Experienced in orchestrating workflows using Airflow, implementing data validation and reconciliation frameworks, and ensuring high data quality in regulated environments. Proven ability to collaborate across engineering, analytics, and compliance teams to deliver reliable and production-grade data solutions.

SKILLS

Programming Languages: Python, SQL, Pyspark, Spark

Cloud Platforms and Tools

- AWS: S3, Glue, IAM, Lambda, CloudWatch
- Azure: Databricks, Data Factory, Synapse
- ETL/ELT Pipelines, Data Lakes (S3, ADLS Gen2), Apache Spark, Apache Kafka, Airflow (DAG design, scheduling, monitoring), dbt (data transformations, modelling, testing)

Infrastructure & DevOps:

- Terraform (Infrastructure as Code)
- CI/CD Pipelines (Azure DevOps / Git-based workflows)

Data Quality & Governance:

- Data Validation & Reconciliation Frameworks
- Data Quality Monitoring & Alerting
- Data Governance & Compliance (Financial Data)

EXPERIENCE

Citibank, Belfast UK

March 2025 - till date

Senior Database Administrator

- Designed and built end-to-end **ETL/ELT pipelines** processing large-scale **financial and trading datasets**
- Orchestrated workflows using **Apache Airflow**, developing **DAGs** for scheduling, dependency management and monitoring
- Built data ingestion pipelines into **S3** and **ADLS Gen2**, integrating structured and unstructured data sources including **Confluent Kafka** topics into **Databricks Delta Lake**
- Implemented **Medallion Architecture (Bronze, Silver, Gold)** in **Databricks**, migrating legacy pipelines to **Delta Lake** with incremental **MERGE** loads ensuring data integrity and full audit trail
- Used **AWS Glue**, **Spark (PySpark)** and **Databricks** for distributed data processing, transformation and large scale **pipeline migration** projects
- Applied **IAM roles** and policies to securely manage access to sensitive financial datasets across **S3** and **Databricks** environments
- Monitored pipelines using **CloudWatch** and **Airflow** alerts, ensuring reliability and quick issue resolution
- Collaborated with compliance and risk teams to support **market surveillance** and **regulatory reporting** systems, delivering **Gold layer** datasets to **Power BI** dashboards

- Designed and built scalable ETL/ELT pipelines using PySpark, Azure Databricks, and Airflow, ensuring
- efficient data processing and reliable workflow orchestration.
- Developed data transformation models using dbt, structuring datasets into reusable layers to support consistent and maintainable analytics.
- Implemented data validation and quality checks using Python and SQL, improving data accuracy and reducing manual data cleaning efforts.
- Ingested batch and streaming data from Kafka, REST APIs, and SQL Server into centralised data lakes (S3 and ADLS Gen2) for downstream use.
- Used Terraform to manage cloud infrastructure, enabling consistent and version-controlled deployments across environments.
- Built and maintained CI/CD pipelines to automate testing, deployment, and monitoring of data workflows, improving delivery speed and reliability.

Bank Of America

Jan 2022 – Dec 2022

Data Engineer

- Worked exclusively with stakeholders across 6 departments to gather requirements, design, build, and execute various data platform initiatives.
- Engineered multiple data pipelines using **Databricks** and **Azure Data Factory (ADF)**, extracting data from over **10** structured and unstructured sources. Additionally, created **triggers** to automate **Data Factory pipelines**, **Dataflows**.
- Orchestrated seamless data ingestion and storage across multiple sources into **Azure Data Lake Storage (ADLS) Gen2**, amplifying data accessibility by **30%**.
- Extracted business and finance data from various sources, including **Azure SQL**, **Snowflake**, **Amazon Redshift**, on-premises SQL Servers, and various web applications, and staged it in ADLS Gen2 using Azure Data Factory (ADF) pipelines.
- Assisted more than **100** end users in troubleshooting data-related issues, performance issues resolving queries, and ensuring the smooth functioning of BI Sols.
- Developed Power BI reports to analyze business trends, improving decision-making efficiency by **20%** and increasing quarterly sales by **15%**.

Accenture

June 2021 – Jan 2023

Data Engineer

- Collaborated with **stakeholders** from various departments, including business users, IT teams, and executives to understand data requirements and translating them into **10+** business dashboards.
- Create and implement an ETL pipeline using **Azure Databricks** and **Data Factory** to automatically extract data from various sources and prepare reporting teams by cleaning data, denormalizing, and organising it.
- Led the successful migration of complex company data from on-premise to cloud using **Databricks** and **Azure Synapse**, Leading to a **25%** increase in performance.
- Established frameworks for **data governance** to guarantee the appropriate management, security, and privacy of data.
- Updating current dashboards, reports, and **dataflows** while focusing on ad hoc tasks to enhance such reports.
- Technically supported internal teams in troubleshooting and resolving **300+** data-related difficulties.

Claro software solutions

July 2020 – May 2021

Data Analyst

- Worked and interpreted diverse client finance and sales datasets to identify trends and insights.
- Delivered bi-monthly reports to upper management, providing strategic insights for decision making, and presented clear data insights to business stakeholders.
- Using Power BI, generated **15+** story-telling dashboards that deliver actionable insights into operational efficiency.
- Published reports to Power BI Service, managed users and user groups, implemented Row-Level Security (RLS), and configured scheduled data refreshes using Gateways for on-premises data sources

EDUCATION

Master's in data science

University of Hertfordshire

Jan 2023 – June 2024

Hatfield, United Kingdom

Bachelor of technology in Computer Science

Gitam University

June 2017 – June 2021

Visakhapatnam, India

CERTIFICATIONS

AWS Cloud Practitioner

DP 700 Fabric Data Engineer Associate

DP 203 – Azure Data Engineer Associate

DP-900 Azure Data Fundamentals

COF-CO2- Snow Pro core Certification

Python Specialization from University of Michigan